

final-eda-part

February 1, 2026

```
[1]: # install all libraries
```

```
!pip install hopsworks shap pandas numpy --upgrade
```

Collecting hopsworks

Downloading hopsworks-4.6.3-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: shap in /usr/local/lib/python3.12/dist-packages (0.50.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)

Collecting pandas

Downloading pandas-3.0.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)

79.5/79.5 kB

2.9 MB/s eta 0:00:00

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)

Collecting numpy

Downloading

numpy-2.4.2-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)

Collecting pyhumps==1.6.1 (from hopsworks)

Downloading pyhumps-1.6.1-py3-none-any.whl.metadata (3.7 kB)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from hopsworks) (2.32.4)

Collecting furl (from hopsworks)

Downloading furl-2.1.4-py2.py3-none-any.whl.metadata (25 kB)

Collecting boto3 (from hopsworks)

Downloading boto3-1.42.39-py3-none-any.whl.metadata (6.8 kB)

Collecting pandas

Downloading pandas-2.3.3-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (91 kB)

91.2/91.2 kB

7.4 MB/s eta 0:00:00

Collecting pyjks (from hopsworks)

Downloading pyjks-20.0.0-py2.py3-none-any.whl.metadata (1.7 kB)

Collecting mock (from hopsworks)

Downloading mock-5.2.0-py3-none-any.whl.metadata (3.1 kB)

Collecting avro==1.12.0 (from hopsworks)

Downloading avro-1.12.0-py2.py3-none-any.whl.metadata (1.7 kB)
 Collecting PyMySQL[rsa] (from hopsworks)
 Downloading pymysql-1.1.2-py3-none-any.whl.metadata (4.3 kB)
 Requirement already satisfied: tzlocal in /usr/local/lib/python3.12/dist-packages (from hopsworks) (5.3.1)
 Requirement already satisfied: fsspec<2025.12.0 in /usr/local/lib/python3.12/dist-packages (from hopsworks) (2025.3.0)
 Collecting retrying (from hopsworks)
 Downloading retrying-1.4.2-py3-none-any.whl.metadata (5.5 kB)
 Collecting hopsworks_aiomysql==0.2.2 (from hopsworks_aiomysql[sa]==0.2.2->hopsworks)
 Downloading hopsworks_aiomysql-0.2.2-py3-none-any.whl.metadata (12 kB)
 Collecting opensearch-py<=2.4.2,>=1.1.0 (from hopsworks)
 Downloading opensearch_py-2.4.2-py2.py3-none-any.whl.metadata (6.8 kB)
 Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from hopsworks) (4.67.1)
 Requirement already satisfied: grpcio<2.0.0,>=1.49.1 in /usr/local/lib/python3.12/dist-packages (from hopsworks) (1.76.0)
 Collecting protobuf<5.0.0,>=4.25.4 (from hopsworks)
 Downloading protobuf-4.25.8-cp37-abi3-manylinux2014_x86_64.whl.metadata (541 bytes)
 Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from hopsworks) (25.0)
 Collecting sqlalchemy<=2.0.41,>=1.3 (from hopsworks_aiomysql[sa]==0.2.2->hopsworks)
 Downloading sqlalchemy-2.0.41-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (9.6 kB)
 Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from shap) (1.16.3)
 Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from shap) (1.6.1)
 Requirement already satisfied: slicer==0.0.8 in /usr/local/lib/python3.12/dist-packages (from shap) (0.0.8)
 Requirement already satisfied: numba>=0.54 in /usr/local/lib/python3.12/dist-packages (from shap) (0.60.0)
 Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from shap) (3.1.2)
 Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from shap) (4.15.0)
 Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
 Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.54->shap) (0.43.0)
 Requirement already satisfied: urllib3>=1.26.18 in

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/usr/local/lib/python3.12/dist-packages (from opensearch-
py<=2.4.2,>=1.1.0->hopsworks) (2.5.0)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from opensearch-py<=2.4.2,>=1.1.0->hopsworks) (1.17.0)
Requirement already satisfied: certifi>=2022.12.07 in
/usr/local/lib/python3.12/dist-packages (from opensearch-
py<=2.4.2,>=1.1.0->hopsworks) (2026.1.4)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->hopsworks) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->hopsworks) (3.11)
Collecting botocore<1.43.0,>=1.42.39 (from boto3->hopsworks)
  Downloading botocore-1.42.39-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3->hopsworks)
  Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3->hopsworks)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Collecting orderedmultidict>=1.0.1 (from furl->hopsworks)
  Downloading orderedmultidict-1.0.2-py2.py3-none-any.whl.metadata (1.2 kB)
Collecting javaobj-py3 (from pyjks->hopsworks)
  Downloading javaobj_py3-0.4.4-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: pyasn1>=0.3.5 in /usr/local/lib/python3.12/dist-
packages (from pyjks->hopsworks) (0.6.2)
Requirement already satisfied: pyasn1-modules in /usr/local/lib/python3.12/dist-
packages (from pyjks->hopsworks) (0.4.2)
Requirement already satisfied: pycryptodomex in /usr/local/lib/python3.12/dist-
packages (from pyjks->hopsworks) (3.23.0)
Collecting twofish (from pyjks->hopsworks)
  Downloading twofish-0.3.0.tar.gz (26 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: cryptography in /usr/local/lib/python3.12/dist-
packages (from PyMySQL[rsa]->hopsworks) (43.0.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-
packages (from scikit-learn->shap) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn->shap) (3.6.0)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-
packages (from
sqlalchemy<=2.0.41,>=1.3->hopsworks_aiomysql[sa]==0.2.2->hopsworks) (3.3.1)
Requirement already satisfied: cffi>=1.12 in /usr/local/lib/python3.12/dist-
packages (from cryptography->PyMySQL[rsa]->hopsworks) (2.0.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-
packages (from cffi>=1.12->cryptography->PyMySQL[rsa]->hopsworks) (3.0)
Downloading hopsworks-4.6.3-py3-none-any.whl (771 kB)
771.5/771.5 kB
19.9 MB/s eta 0:00:00
Downloading avro-1.12.0-py2.py3-none-any.whl (124 kB)
124.2/124.2 kB

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9.3 MB/s eta 0:00:00
Downloading hopsworks_aiomysql-0.2.2-py3-none-any.whl (44 kB)
44.2/44.2 kB

3.2 MB/s eta 0:00:00
Downloading pyhumps-1.6.1-py3-none-any.whl (5.0 kB)
Downloading
pandas-2.3.3-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (12.4
MB)
12.4/12.4 MB

73.3 MB/s eta 0:00:00
Downloading opensearch_py-2.4.2-py2.py3-none-any.whl (258 kB)
258.6/258.6 kB

22.6 MB/s eta 0:00:00
Downloading protobuf-4.25.8-cp37-abi3-manylinux2014_x86_64.whl (294 kB)
294.9/294.9 kB

25.1 MB/s eta 0:00:00
Downloading boto3-1.42.39-py3-none-any.whl (140 kB)
140.6/140.6 kB

13.7 MB/s eta 0:00:00
Downloading furl-2.1.4-py2.py3-none-any.whl (27 kB)
Downloading mock-5.2.0-py3-none-any.whl (31 kB)
Downloading pyjks-20.0.0-py2.py3-none-any.whl (45 kB)
45.3/45.3 kB

4.1 MB/s eta 0:00:00
Downloading retrying-1.4.2-py3-none-any.whl (10 kB)
Downloading botocore-1.42.39-py3-none-any.whl (14.6 MB)
14.6/14.6 MB

63.2 MB/s eta 0:00:00
Downloading jmespath-1.1.0-py3-none-any.whl (20 kB)
Downloading orderedmultidict-1.0.2-py2.py3-none-any.whl (11 kB)
Downloading pymysql-1.1.2-py3-none-any.whl (45 kB)
45.3/45.3 kB

3.9 MB/s eta 0:00:00
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
86.8/86.8 kB

8.4 MB/s eta 0:00:00
Downloading
sqlalchemy-2.0.41-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(3.3 MB)
3.3/3.3 MB

83.7 MB/s eta 0:00:00
Downloading javaobj_py3-0.4.4-py2.py3-none-any.whl (57 kB)
57.1/57.1 kB

4.3 MB/s eta 0:00:00
Building wheels for collected packages: twofish
  Building wheel for twofish (setup.py) ... done
  Created wheel for twofish: filename=twofish-0.3.0-cp312-cp312-linux_x86_64.whl
size=24321

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sha256=493be9443102f55be461efd85c5a5dbc410b621c5d356d087165272772bd7d73
  Stored in directory: /root/.cache/pip/wheels/fa/81/02/abf836d4acb19a3de48f6bfd
738cb9bcb762978b835bca2faa
Successfully built twofish
Installing collected packages: twofish, pyhumps, javaobj-py3, sqlalchemy,
retrying, PyMySQL, protobuf, orderedmultidict, mock, jmespath, avro, pyjks,
pandas, opensearch-py, hopsworks_aiomysql, furl, botocore, s3transfer, boto3,
hopsworks
  Attempting uninstall: sqlalchemy
    Found existing installation: SQLAlchemy 2.0.46
    Uninstalling SQLAlchemy-2.0.46:
      Successfully uninstalled SQLAlchemy-2.0.46
  Attempting uninstall: protobuf
    Found existing installation: protobuf 5.29.5
    Uninstalling protobuf-5.29.5:
      Successfully uninstalled protobuf-5.29.5
  Attempting uninstall: pandas
    Found existing installation: pandas 2.2.2
    Uninstalling pandas-2.2.2:
      Successfully uninstalled pandas-2.2.2
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.

google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.3.3 which is
incompatible.

grain 0.2.15 requires protobuf>=5.28.3, but you have protobuf 4.25.8 which is
incompatible.

ydf 0.14.0 requires protobuf<7.0.0,>=5.29.1, but you have protobuf 4.25.8 which
is incompatible.

opentelemetry-proto 1.37.0 requires protobuf<7.0,>=5.0, but you have protobuf
4.25.8 which is incompatible.

grpcio-status 1.71.2 requires protobuf<6.0dev,>=5.26.1, but you have protobuf
4.25.8 which is incompatible.

Successfully installed PyMySQL-1.1.2 avro-1.12.0 boto3-1.42.39
botocore-1.42.39 furl-2.1.4 hopsworks-4.6.3 hopsworks_aiomysql-0.2.2 javaobj-
py3-0.4.4 jmespath-1.1.0 mock-5.2.0 opensearch-py-2.4.2 orderedmultidict-1.0.2
pandas-2.3.3 protobuf-4.25.8 pyhumps-1.6.1 pyjks-20.0.0 retrying-1.4.2
s3transfer-0.16.0 sqlalchemy-2.0.41 twofish-0.3.0

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```
[1]: !pip install hopsworks==4.2.*
```

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Collecting hopsworks==4.2.*
  Downloading hopsworks-4.2.10-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: pyhumps==1.6.1 in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (1.6.1)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (2.32.4)
Requirement already satisfied: furl in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (2.1.4)
Requirement already satisfied: boto3 in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (1.42.39)
Collecting pandas<2.2.0 (from hopsworks==4.2.*)
  Downloading
pandas-2.1.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (18 kB)
Requirement already satisfied: pyjks in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (20.0.0)
Requirement already satisfied: mock in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (5.2.0)
Collecting avro==1.11.3 (from hopsworks==4.2.*)
  Downloading avro-1.11.3.tar.gz (90 kB)
                                90.6/90.6 kB
3.3 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: sqlalchemy in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (2.0.41)
Requirement already satisfied: PyMySQL[rsa] in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (1.1.2)
Requirement already satisfied: tzlocal in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (5.3.1)
Requirement already satisfied: fsspec<2025.12.0 in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (2025.3.0)
Requirement already satisfied: retrying in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (1.4.2)
Collecting hopsworks_aiomysql==0.2.1 (from hopsworks_aiomysql[sa]==0.2.1->hopsworks==4.2.*)
  Downloading hopsworks_aiomysql-0.2.1-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: opensearch-py<=2.4.2,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (2.4.2)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (4.67.1)
Requirement already satisfied: grpcio<2.0.0,>=1.49.1 in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (1.76.0)
Requirement already satisfied: protobuf<5.0.0,>=4.25.4 in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (4.25.8)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from hopsworks==4.2.*) (25.0)

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Collecting sqlalchemy (from hopsworks==4.2.*)
  Downloading SQLAlchemy-2.0.29-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (9.6 kB)
Requirement already satisfied: typing-extensions~=4.12 in
/usr/local/lib/python3.12/dist-packages (from
grpcio<2.0.0,>=1.49.1->hopsworks==4.2.*) (4.15.0)
Requirement already satisfied: urllib3>=1.26.18 in
/usr/local/lib/python3.12/dist-packages (from opensearch-
py<=2.4.2,>=1.1.0->hopsworks==4.2.*) (2.5.0)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from opensearch-py<=2.4.2,>=1.1.0->hopsworks==4.2.*) (1.17.0)
Requirement already satisfied: python-dateutil in
/usr/local/lib/python3.12/dist-packages (from opensearch-
py<=2.4.2,>=1.1.0->hopsworks==4.2.*) (2.9.0.post0)
Requirement already satisfied: certifi>=2022.12.07 in
/usr/local/lib/python3.12/dist-packages (from opensearch-
py<=2.4.2,>=1.1.0->hopsworks==4.2.*) (2026.1.4)
Collecting numpy<2,>=1.26.0 (from pandas<2.2.0->hopsworks==4.2.*)
  Downloading
numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)

61.0/61.0 kB

3.4 MB/s eta 0:00:00
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas<2.2.0->hopsworks==4.2.*)
(2025.2)
Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.12/dist-
packages (from pandas<2.2.0->hopsworks==4.2.*) (2025.3)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->hopsworks==4.2.*)
(3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->hopsworks==4.2.*) (3.11)
Requirement already satisfied: greenlet!=0.4.17 in
/usr/local/lib/python3.12/dist-packages (from sqlalchemy->hopsworks==4.2.*)
(3.3.1)
Requirement already satisfied: botocore<1.43.0,>=1.42.39 in
/usr/local/lib/python3.12/dist-packages (from boto3->hopsworks==4.2.*) (1.42.39)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in
/usr/local/lib/python3.12/dist-packages (from boto3->hopsworks==4.2.*) (1.1.0)
Requirement already satisfied: s3transfer<0.17.0,>=0.16.0 in
/usr/local/lib/python3.12/dist-packages (from boto3->hopsworks==4.2.*) (0.16.0)
Requirement already satisfied: orderedmultidict>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from furl->hopsworks==4.2.*) (1.0.2)
Requirement already satisfied: javaobj-py3 in /usr/local/lib/python3.12/dist-
packages (from pyjks->hopsworks==4.2.*) (0.4.4)
Requirement already satisfied: pyasn1>=0.3.5 in /usr/local/lib/python3.12/dist-
packages (from pyjks->hopsworks==4.2.*) (0.6.2)

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Requirement already satisfied: pyasn1-modules in /usr/local/lib/python3.12/dist-packages (from pyjks->hopsworks==4.2.*) (0.4.2)
Requirement already satisfied: pycryptodomex in /usr/local/lib/python3.12/dist-packages (from pyjks->hopsworks==4.2.*) (3.23.0)
Requirement already satisfied: twofish in /usr/local/lib/python3.12/dist-packages (from pyjks->hopsworks==4.2.*) (0.3.0)
Requirement already satisfied: cryptography in /usr/local/lib/python3.12/dist-packages (from PyMySQL[rsa]->hopsworks==4.2.*) (43.0.3)
Requirement already satisfied: cffi>=1.12 in /usr/local/lib/python3.12/dist-packages (from cryptography->PyMySQL[rsa]->hopsworks==4.2.*) (2.0.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.12->cryptography->PyMySQL[rsa]->hopsworks==4.2.*) (3.0)
Downloading hopsworks-4.2.10-py3-none-any.whl (665 kB)

665.0/665.0 kB

16.9 MB/s eta 0:00:00

Downloading hopsworks_aiomysql-0.2.1-py3-none-any.whl (44 kB)

44.2/44.2 kB

3.1 MB/s eta 0:00:00

Downloading

pandas-2.1.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (11.7 MB)

11.7/11.7 MB

78.0 MB/s eta 0:00:00

Downloading

SQLAlchemy-2.0.29-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.2 MB)

3.2/3.2 MB

76.9 MB/s eta 0:00:00

Downloading

numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.0 MB)

18.0/18.0 MB

58.9 MB/s eta 0:00:00

Building wheels for collected packages: avro

Building wheel for avro (pyproject.toml) ... done

Created wheel for avro: filename=avro-1.11.3-py2.py3-none-any.whl size=123962 sha256=4c3808b12469988318e0f1dc88e56b4329fee9ec517c9ebad07113de072ec78e

Stored in directory: /root/.cache/pip/wheels/3a/82/d3/8eb96fda033c7f1661086e2f8afb13f04817886d28b12f1e72

Successfully built avro

Installing collected packages: sqlalchemy, numpy, hopsworks_aiomysql, avro, pandas, hopsworks

Attempting uninstall: sqlalchemy

Found existing installation: SQLAlchemy 2.0.41

Uninstalling SQLAlchemy-2.0.41:

Successfully uninstalled SQLAlchemy-2.0.41

Attempting uninstall: numpy

Found existing installation: numpy 2.0.2

```
Uninstalling numpy-2.0.2:
  Successfully uninstalled numpy-2.0.2
Attempting uninstall: hopsworks_aiomysql
  Found existing installation: hopsworks-aiomysql 0.2.2
  Uninstalling hopsworks-aiomysql-0.2.2:
    Successfully uninstalled hopsworks-aiomysql-0.2.2
Attempting uninstall: avro
  Found existing installation: avro 1.12.0
  Uninstalling avro-1.12.0:
    Successfully uninstalled avro-1.12.0
Attempting uninstall: pandas
  Found existing installation: pandas 2.3.3
  Uninstalling pandas-2.3.3:
    Successfully uninstalled pandas-2.3.3
Attempting uninstall: hopsworks
  Found existing installation: hopsworks 4.6.3
  Uninstalling hopsworks-4.6.3:
    Successfully uninstalled hopsworks-4.6.3
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.1.4 which is incompatible.

opencv-python-headless 4.13.0.90 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.

rasterio 1.5.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.

tobler 0.13.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

tobler 0.13.0 requires pandas>=2.2, but you have pandas 2.1.4 which is incompatible.

grain 0.2.15 requires protobuf>=5.28.3, but you have protobuf 4.25.8 which is incompatible.

opencv-contrib-python 4.13.0.90 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.

pytensor 2.37.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

jax 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

ydf 0.14.0 requires protobuf<7.0.0,>=5.29.1, but you have protobuf 4.25.8 which is incompatible.

opencv-python 4.13.0.90 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.

jaxlib 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

mizani 0.13.5 requires pandas>=2.2.0, but you have pandas 2.1.4 which is incompatible.

plotnine 0.14.5 requires pandas>=2.2.0, but you have pandas 2.1.4 which is incompatible.

xarray 2025.12.0 requires pandas>=2.2, but you have pandas 2.1.4 which is incompatible.

shap 0.50.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.

numpy-1.26.4 pandas-2.1.4 sqlalchemy-2.0.29

```
[1]: import hopsworks
import joblib
import shap
import matplotlib.pyplot as plt
import pandas as pd
import os
from google.colab import userdata
import seaborn as sns
import numpy as np
```

```
[2]: import hopsworks

project = hopsworks.login()
fs = project.get_feature_store()
```

Copy your Api Key (first register/login):
<https://c.app.hopsworks.ai/account/api/generated>

Paste it here:

Logged in to project, explore it here <https://c.app.hopsworks.ai:443/p/1335449>

```
[3]: # Get your Feature Group
aqi_fg = fs.get_feature_group(name="islamabad_aqi_v12", version=5)
```

```
[4]: raw_df = aqi_fg.read()
```

Finished: Reading data from Hopsworks, using Hopsworks Feature Query Service (0.61s)

```
[ ]: sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (12,5)
```

```
[15]: print("Data Preview (last rows):")
display(raw_df.tail(5))
```

Data Preview (last rows):

	datetime	city	aqi	pm2_5	no2	so2	\
2965	2026-02-01 07:00:00+00:00	Islamabad	5.0	157.85	4.69	5.66	
2966	2026-02-01 08:00:00+00:00	Islamabad	5.0	114.61	2.50	4.19	
2967	2026-02-01 09:00:00+00:00	Islamabad	5.0	95.27	2.15	3.44	
2968	2026-02-01 10:00:00+00:00	Islamabad	5.0	89.72	3.34	2.92	
2969	2026-02-01 11:00:00+00:00	Islamabad	5.0	93.93	5.90	2.63	

temperature humidity wind_speed hour weekday month aqi_lag_1 \

2965	15.4	54.0	7.4	7.0	6.0	2.0	5.0
2966	16.0	52.0	9.3	8.0	6.0	2.0	5.0
2967	16.5	51.0	11.1	9.0	6.0	2.0	5.0
2968	16.1	52.0	11.1	10.0	6.0	2.0	5.0
2969	15.7	54.0	11.1	11.0	6.0	2.0	5.0

	pm2_5_rolling_6h	wind_stagnant
2965	237.668333	0.0
2966	227.121667	0.0
2967	208.183333	0.0
2968	184.538333	0.0
2969	159.160000	0.0

```
[ ]: print("Data Preview (First 5 rows):")
display(raw_df.head())
```

Data Preview (First 5 rows):

	datetime	city	aqi	pm2_5	no2	so2	temperature	\
0	2025-12-21 04:00:00+00:00	Islamabad	5.0	392.74	15.86	1.80	14.2	
1	2025-10-27 18:00:00+00:00	Islamabad	5.0	105.79	25.06	7.04	18.6	
2	2025-12-26 04:00:00+00:00	Islamabad	5.0	157.86	29.04	6.19	10.1	
3	2025-10-26 21:00:00+00:00	Islamabad	5.0	111.92	16.38	6.15	17.1	
4	2026-01-04 12:00:00+00:00	Islamabad	5.0	111.82	13.58	3.56	14.5	

	humidity	wind_speed	hour	weekday	month	aqi_lag_1	pm2_5_rolling_6h	\
0	71.0	9.3	4.0	6.0	12.0	5.0	393.598333	
1	69.0	5.5	18.0	0.0	10.0	5.0	89.588333	
2	67.0	1.8	4.0	4.0	12.0	5.0	138.853333	
3	67.0	1.8	21.0	6.0	10.0	5.0	96.740000	
4	62.0	9.3	12.0	6.0	1.0	5.0	116.861667	

	wind_stagnant
0	0.0
1	0.0
2	1.0
3	1.0
4	0.0

```
[16]: cols_to_show = ['datetime', 'aqi', 'pm2_5', 'hour', 'weekday', 'aqi_lag_1']
print("Strategic Data Preview:")
display(raw_df[cols_to_show].tail(10))
```

Strategic Data Preview:

	datetime	aqi	pm2_5	hour	weekday	aqi_lag_1
2960	2026-02-01 00:00:00+00:00	5.0	217.15	0.0	6.0	5.0
2961	2026-02-01 04:00:00+00:00	5.0	241.99	4.0	6.0	5.0
2962	2026-02-01 03:00:00+00:00	5.0	237.14	3.0	6.0	5.0

2963	2026-02-01 05:00:00+00:00	5.0	245.25	5.0	6.0	5.0
2964	2026-02-01 06:00:00+00:00	5.0	252.26	6.0	6.0	5.0
2965	2026-02-01 07:00:00+00:00	5.0	157.85	7.0	6.0	5.0
2966	2026-02-01 08:00:00+00:00	5.0	114.61	8.0	6.0	5.0
2967	2026-02-01 09:00:00+00:00	5.0	95.27	9.0	6.0	5.0
2968	2026-02-01 10:00:00+00:00	5.0	89.72	10.0	6.0	5.0
2969	2026-02-01 11:00:00+00:00	5.0	93.93	11.0	6.0	5.0

```
[17]: # Quick check for total rows and data types
print("\nDataset Info:")
raw_df.info()
```

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2970 entries, 0 to 2969
```

```
Data columns (total 15 columns):
```

#	Column	Non-Null Count	Dtype
0	datetime	2970 non-null	datetime64[us, Etc/UTC]
1	city	2970 non-null	object
2	aqi	2970 non-null	float64
3	pm2_5	2970 non-null	float64
4	no2	2970 non-null	float64
5	so2	2970 non-null	float64
6	temperature	2970 non-null	float64
7	humidity	2970 non-null	float64
8	wind_speed	2970 non-null	float64
9	hour	2970 non-null	float64
10	weekday	2970 non-null	float64
11	month	2970 non-null	float64
12	aqi_lag_1	2970 non-null	float64
13	pm2_5_rolling_6h	2970 non-null	float64
14	wind_stagnant	2970 non-null	float64

```
dtypes: datetime64[us, Etc/UTC](1), float64(13), object(1)
```

```
memory usage: 348.2+ KB
```

```
[ ]: print("Null values per column:")
print(raw_df.isnull().sum())
```

Null values per column:

datetime	0
city	0
aqi	0
pm2_5	0
no2	0
so2	0
temp	0

```

humidity          0
wind_speed        0
hour              0
weekday           0
month             0
aqi_lag_1         0
pm2_5_rolling_6h  0
wind_stagnant     0
dtype: int64

```

```
[ ]: raw_df.isnull().sum()
```

EDA Graphs

Daily Average AQI Trend

```
[ ]: import plotly.express as px

# --- GRAPH 1: (Daily Average AQI) ---
df_daily = raw_df.resample('D', on='datetime').aqi.mean().reset_index()

fig1 = px.bar(df_daily, x='datetime', y='aqi',
              title='Islamabad Air Quality: Last Few Days',
              labels={'aqi': 'Pollution Level (1=Best, 5=Worst)', 'datetime': 'Date'},
              color='aqi',
              color_continuous_scale=['green', 'yellow', 'orange', 'red', 'purple'])
fig1.show()
```

AQI Category Distribution

```
[ ]: # --- GRAPH 2: AQI Scoreboard (Pie Chart) ---
# Air Quality
status_map = {1: "Good (clean) ", 2: "Fair", 3: "Moderate", 4: "Poor", 5: "Very Poor (Dangerous)"}
raw_df['Condition'] = raw_df['aqi'].round().map(status_map)

fig2 = px.pie(raw_df, names='Condition',
              title='Overall Air Quality Distribution in Islamabad',
              color_discrete_map={'Good (clean)': 'green', 'Very Poor (Dangerous)': 'purple', 'Poor': 'red'})
fig2.show()
```

Temperature vs AQI Relationship

```
[ ]: # --- GRAPH 3: (Temperature vs AQI) ---
# Is pollution increase in cold?
```

```
fig3 = px.box(raw_df, x='aqi', y='temperature',
              title='Relation between Cold Weather and Pollution',
              labels={'aqi': 'Pollution Level', 'temperature': 'Temperature_
↳(°C)'},
              color='aqi')
fig3.show()
```

Weather Factors Impacting AQI

```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# 1. AQI ke saath baaqi cheezon ka talluq (Correlation) nikalna
# Hum sirf un columns ko le rahe hain jo layman samajh sakay
features = ['aqi', 'temperature', 'humidity', 'wind_speed', 'pm2_5']
corr_data = raw_df[features].corr()['aqi'].drop('aqi').reset_index()
corr_data.columns = ['Weather Factor', 'Impact_Score']

# Score ko absolute (positive) kar dete hain taake sirf taqat nazar aaye
corr_data['Impact_Score'] = corr_data['Impact_Score'].abs()
corr_data = corr_data.sort_values(by='Impact_Score', ascending=False)

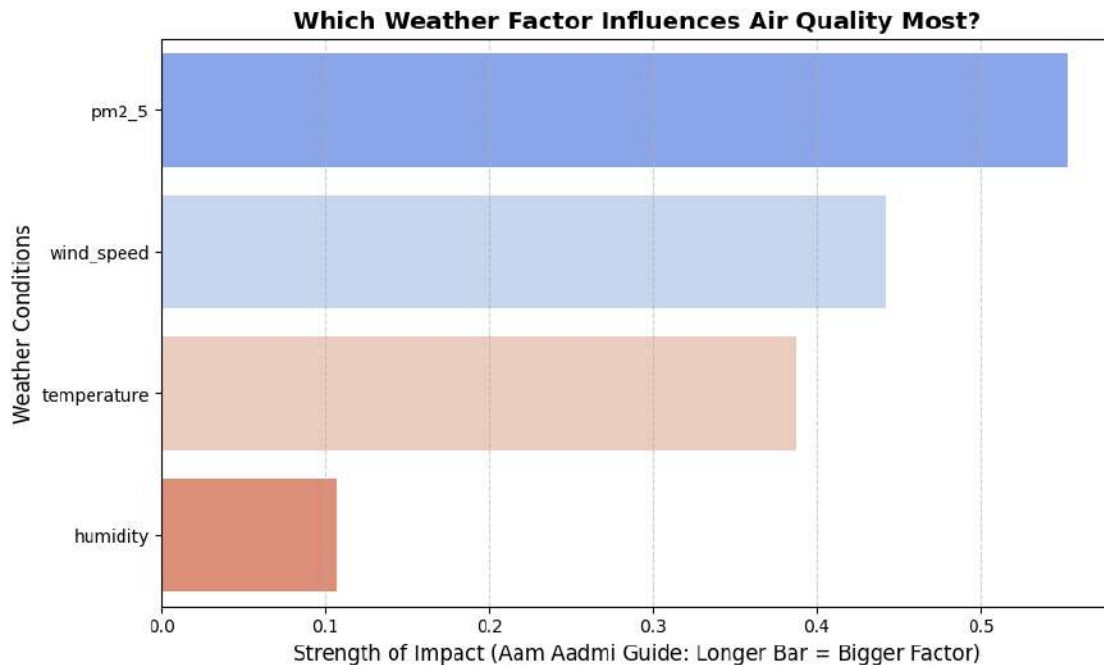
# 2. Simple Bar Plot
plt.figure(figsize=(10, 6))
# 'Impact_Score' jitna zyada hoga, bar utni lambi hogi
sns.barplot(x='Impact_Score', y='Weather Factor', data=corr_data,
↳palette='coolwarm')

plt.title('Which Weather Factor Influences Air Quality Most?', fontsize=14,
↳fontweight='bold')
plt.xlabel('Strength of Impact (Aam Aadmi Guide: Longer Bar = Bigger Factor)',
↳fontsize=12)
plt.ylabel('Weather Conditions', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.6)

plt.show()
```

/tmp/ipython-input-1062513653.py:18: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.



Monthly Average AQI Trends

```
[ ]: import plotly.express as px

# 1. Data ko pehle sort karein aur mahine ka naam nikalein
raw_df['datetime'] = pd.to_datetime(raw_df['datetime'])
df_sorted = raw_df.sort_values('datetime')
df_sorted['Month'] = df_sorted['datetime'].dt.strftime('%B %Y')

# 2. Monthly average nikalein
# Hum 'numeric_only=True' use kar rahe hain taake sirf numbers ka mean niklay
df_monthly = df_sorted.groupby('Month', sort=False)['aqi'].mean().reset_index()

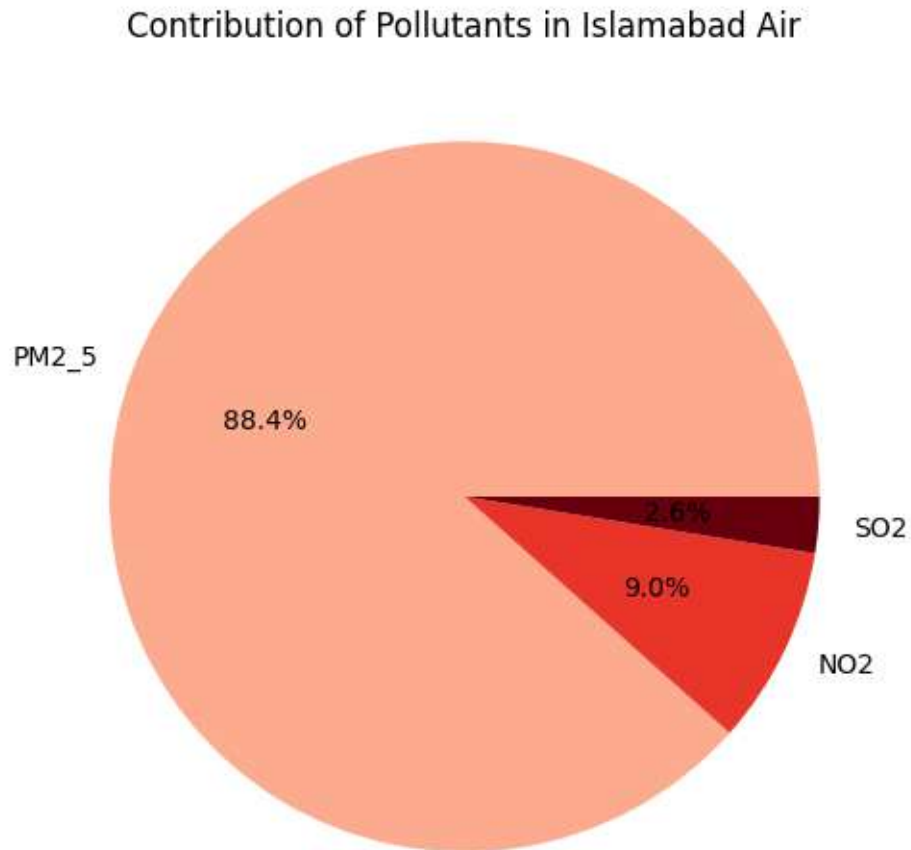
# 3. Simple Bar Graph
fig = px.bar(df_monthly,
             x='Month',
             y='aqi',
             title='Islamabad: Average Pollution Level by Month',
             labels={'aqi': 'Average AQI (1-5)', 'Month': 'Month'},
             color='aqi',
             color_continuous_scale='RdYlGn_r', # Green to Red scale
             text_auto='.1f') # Bars ke upar value dikhane ke liye

fig.update_layout(xaxis_tickangle=-45)
fig.show()
```

Pollutant Contribution to AQI

```
[ ]: pollutants = ['pm2_5', 'no2', 'so2']
avg_values = raw_df[pollutants].mean()

plt.figure(figsize=(6, 6))
plt.pie(avg_values, labels=[p.upper() for p in pollutants], autopct='%1.1f%%',
        colors=plt.cm.Reds(np.linspace(0.3, 1, len(pollutants))))
plt.title('Contribution of Pollutants in Islamabad Air')
plt.savefig('Contribution of Pollutants in Islamabad Air')
plt.show()
```



Distribution of Key Pollutants

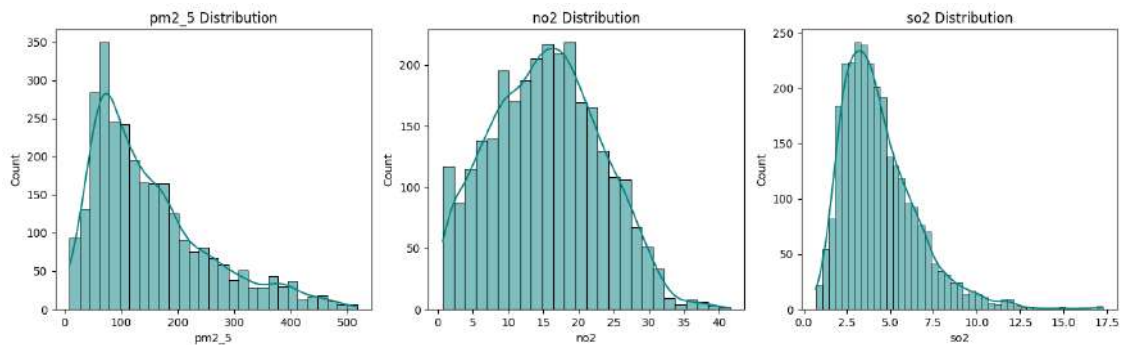
```
[ ]: # Pollutant Distribution (PM2.5, PM10, CO, etc.)

pollutants = ['pm2_5', 'no2', 'so2']
```

```

plt.figure(figsize=(14,8))
for i, col in enumerate(pollutants):
    plt.subplot(2, 3, i+1)
    sns.histplot(raw_df[col], kde=True, color='teal')
    plt.title(f'{col} Distribution')
plt.tight_layout()
plt.savefig("Pollutants Distribution.png")
plt.show()

```



Most Critical Pollutant Analysis

```

[ ]: # Define your pollutants
pollutants = ['pm2_5', 'no2', 'so2']

# Calculate average concentration of each pollutant
avg_concentrations = raw_df[pollutants].mean().sort_values(ascending=False)

# Identify the most critical one
most_critical = avg_concentrations.idxmax()
most_value = avg_concentrations.max()

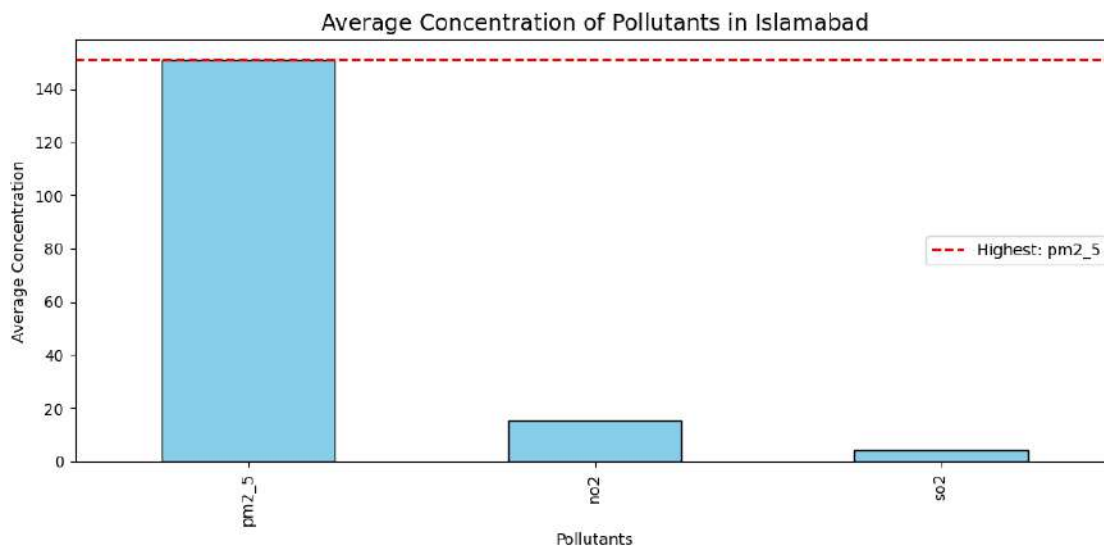
# Print result
print('='*60)
print(f"Most critical pollutant in Islamabad: {most_critical} (Average_↵
↵concentration: {most_value:.2f})")
print('='*60)

# Plot
plt.figure(figsize=(10, 5))
avg_concentrations.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Average Concentration of Pollutants in Islamabad', fontsize=14)
plt.ylabel('Average Concentration')
plt.xlabel('Pollutants')

```

```
plt.axhline(y=most_value, color='red', linestyle='--', label=f'Highest: {most_critical}')
plt.legend()
plt.tight_layout()
plt.savefig('Average Concentration of Pollutants in Islamabad.png')
plt.show()
```

```
=====
Most critical pollutant in Islamabad: pm2_5 (Average concentration: 150.97)
=====
```



Correlation Analysis of AQI and Features

```
[ ]: import numpy as np
def correlation_analysis(df):
    """
    Perform correlation analysis between all features (including AQI),
    print strongest correlated pairs, show heatmap, and barplot of AQI
    correlations.
    """
    print("\n POLLUTANT CORRELATION ANALYSIS")
    print("="*60)

    # Select only numeric columns for correlation analysis
    df_numeric = df.select_dtypes(include=np.number)

    # Compute correlation matrix
    corr = df_numeric.corr()
```

```

# Print strongest correlation pairs
print("\n STRONGEST CORRELATIONS (|corr| > 0.7):")
print("-" * 40)
strong = []
for i in range(len(corr.columns)):
    for j in range(i+1, len(corr.columns)):
        val = corr.iloc[i, j]
        if abs(val) > 0.7:
            strong.append((corr.index[i], corr.columns[j], val))
    for var1, var2, val in sorted(strong, key=lambda x: abs(x[2]),
↪reverse=True):
        print(f"{var1}    {var2}: {val:.3f}")

# Heatmap
print("\n Displaying Correlation Heatmap...")
plt.figure(figsize=(12, 8))
sns.heatmap(corr, annot=True, cmap='Blues', fmt=".2f", square=True)
plt.title("Correlation Heatmap of AQI & Pollutants", fontsize=14)
plt.xticks(rotation=45)
plt.yticks(rotation=0)
plt.tight_layout()
plt.savefig("Correlation Heatmap of AQI & Pollutants.png")
plt.show()

# AQI Correlation Bar Plot
if 'aqi' in corr.columns: # Changed 'AQI' to 'aqi'
    print("\n Correlation of Features with AQI:")
    corr_with_aqi = corr['aqi'].sort_values(ascending=False) # Changed
↪'AQI' to 'aqi'
    print(corr_with_aqi)

    plt.figure(figsize=(10, 6))
    corr_with_aqi.drop('aqi').plot(kind='barh', color='skyblue') # Changed
↪'AQI' to 'aqi'
    plt.title(' Feature Correlation with AQI', fontsize=14)
    plt.xlabel('Correlation Coefficient')
    plt.gca().invert_yaxis()
    plt.grid(axis='x', linestyle='--', alpha=0.7)
    plt.tight_layout()
    plt.savefig("Feature Correlation with AQI.png")
    plt.show()
else:
    print(" 'aqi' column not found in the dataset.")

# Call the function with raw_df
correlation_analysis(raw_df)

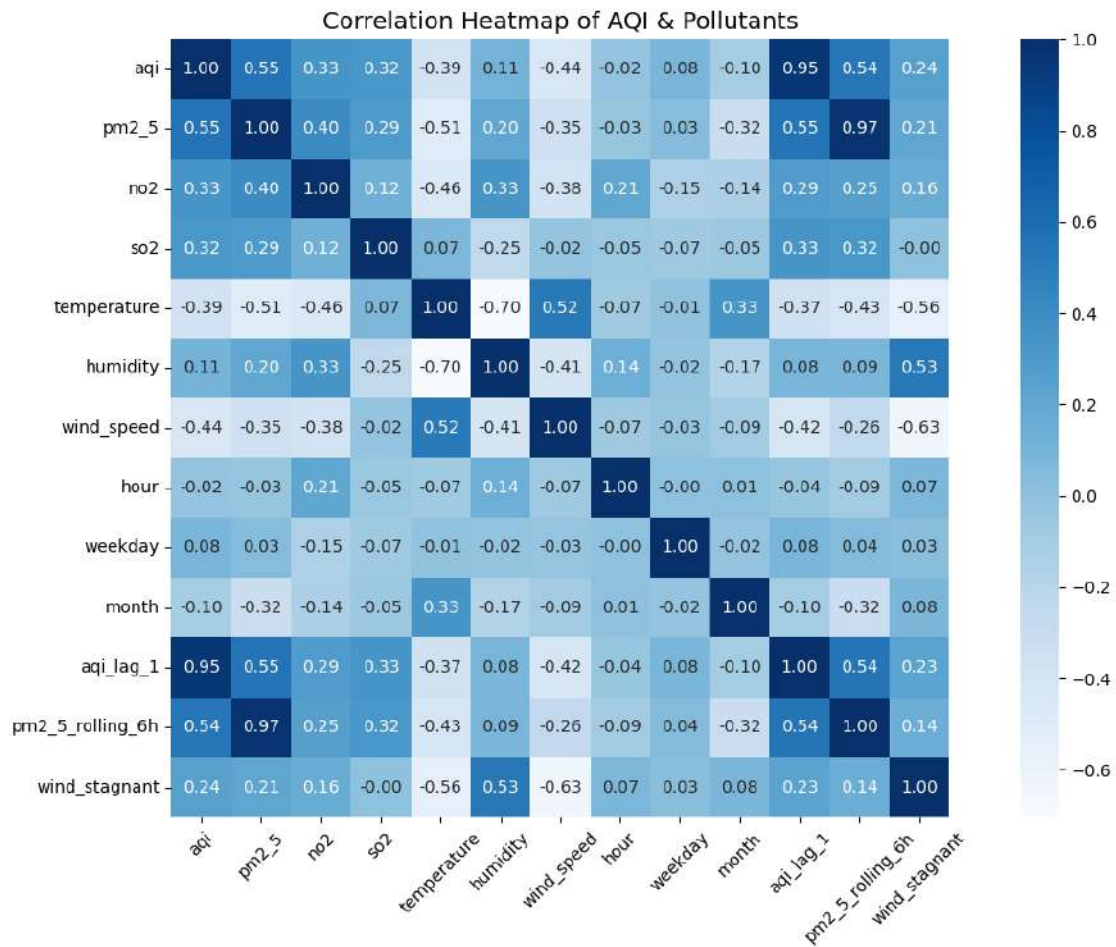
```

POLLUTANT CORRELATION ANALYSIS

STRONGEST CORRELATIONS ($|\text{corr}| > 0.7$):

pm2_5 pm2_5_rolling_6h: 0.967
 aqi aqi_lag_1: 0.952
 temperature humidity: -0.703

Displaying Correlation Heatmap...



Correlation of Features with AQI:

aqi 1.000000
 aqi_lag_1 0.952167
 pm2_5 0.553317
 pm2_5_rolling_6h 0.537691

```

no2          0.333047
so2          0.318288
wind_stagnant 0.242048
humidity     0.107636
weekday      0.082832
hour         -0.018024
month        -0.099265
temperature  -0.387629
wind_speed   -0.442380
Name: aqi, dtype: float64

```

/tmp/ipython-input-563425939.py:51: UserWarning:

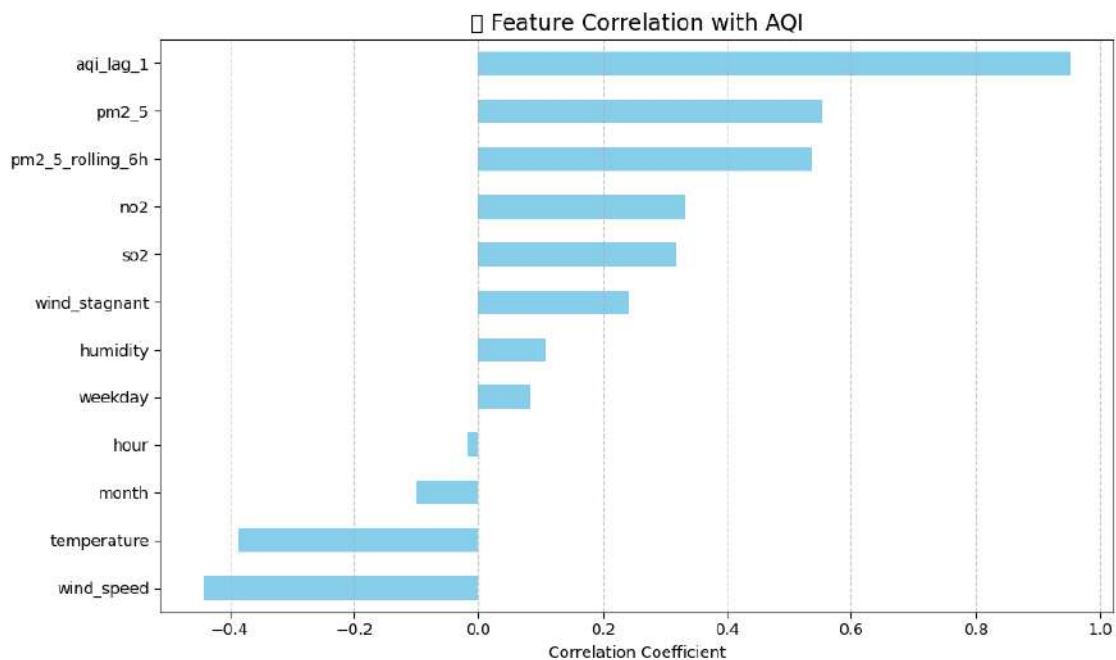
Glyph 128269 (\N{LEFT-POINTING MAGNIFYING GLASS}) missing from font(s) DejaVu Sans.

/tmp/ipython-input-563425939.py:52: UserWarning:

Glyph 128269 (\N{LEFT-POINTING MAGNIFYING GLASS}) missing from font(s) DejaVu Sans.

/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151:
UserWarning:

Glyph 128269 (\N{LEFT-POINTING MAGNIFYING GLASS}) missing from font(s) DejaVu Sans.



SHAP Analysis

```
[14]: # Colab-friendly SHAP Analysis for AQI Model

import hopsworks
import joblib
import shap
import matplotlib.pyplot as plt
import os
from dotenv import load_dotenv # Added for environment variables

load_dotenv() # Load environment variables

# --- 1 Initialize Hopsworks Connection ---
# Ensure project, fs, mr are available
project = hopsworks.login(api_key_value=os.getenv("HOPSWORKS_KEY")) # Login to Hopsworks
fs = project.get_feature_store() # Get feature store
mr = project.get_model_registry() # Get model registry

# --- 2 Download & load best model ---
best_model_registry = mr.get_model("best_islamabad_aqi_model")
model_dir = best_model_registry.download() # 'model_dir' will now hold the path to the downloaded directory

# Find the actual .pkl file within the downloaded directory
model_files = [f for f in os.listdir(model_dir) if f.endswith('.pkl')]
if not model_files:
    raise FileNotFoundError(f"No .pkl model file found in the downloaded directory: {model_dir}")
model_filename = model_files[0]
model_path = os.path.join(model_dir, model_filename) # Correctly construct the path to the model file

best_model = joblib.load(model_path)
print(" Best model loaded:", best_model)

# --- 3 Fetch test features from Feature View ---
fv = fs.get_feature_view(name="islamabad_aqi_viewss", version=1)
X_train, X_test, y_train, y_test = fv.train_test_split(test_size=0.2)

# --- 4 Clean features for SHAP ---
cols_to_drop = ['aqi', 'pm2_5', 'city', 'datetime', 'timestamp', 'no2', 'so2']
```

```

def clean_features(df):
    df = df.drop(columns=[c for c in cols_to_drop if c in df.columns])
    return df.select_dtypes(include=['number']).fillna(0)

X_test = clean_features(X_test)
print(f" Features ready: {list(X_test.columns)}")

# --- 5 Initialize SHAP Explainer ---
explainer = shap.TreeExplainer(best_model)
shap_values = explainer.shap_values(X_test)

# --- 6 Global Feature Importance (Bar Plot) ---
shap.summary_plot(shap_values, X_test, plot_type="bar", show=True)
plt.savefig("shap_summary_bar.png", dpi=300)

# --- 7 Detailed Summary Plot (Impact of feature values) ---
shap.summary_plot(shap_values, X_test, show=True)
plt.savefig("shap_summary_detailed.png", dpi=300)

# --- 8 Dependence Plot for a Key Feature ---
shap.dependence_plot("pm2_5_rolling_6h", shap_values, X_test)
plt.savefig("shap_dependence_pm2_5.png", dpi=300)

# --- 9 Force Plot for a Single Prediction (local explanation) ---
shap.initjs() # Works in Colab for interactive JS plots
sample = X_test.iloc[0,:]
shap.force_plot(explainer.expected_value, shap_values[0,:], sample)
# Force plot will show inline in Colab

print(" SHAP analysis completed successfully in Colab!")

```

Connection closed.

Logged in to project, explore it here <https://c.app.hopsworks.ai:443/p/1335449>

VersionWarning: No version provided for getting model

`best\_islamabad\_aqi\_model`, defaulting to `1`.

Downloading: 0.000%| | 0/161881 elapsed<00:00 remaining<?

```

Best model loaded: XGBRegressor(base_score=None, booster=None, callbacks=None,
                                colsample_bylevel=None, colsample_bynode=None,
                                colsample_bytree=None, device=None, early_stopping_rounds=None,
                                enable_categorical=False, eval_metric=None, feature_types=None,
                                feature_weights=None, gamma=None, grow_policy=None,
                                importance_type=None, interaction_constraints=None,
                                learning_rate=0.05, max_bin=None, max_cat_threshold=None,
                                max_cat_to_onehot=None, max_delta_step=None, max_depth=4,

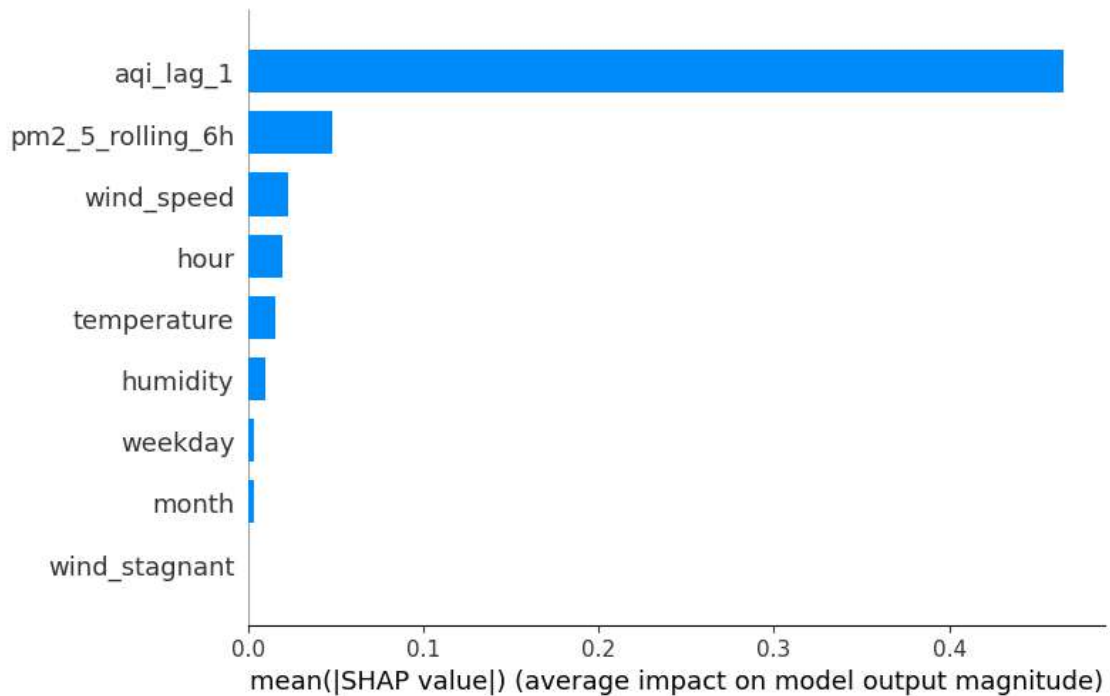
```

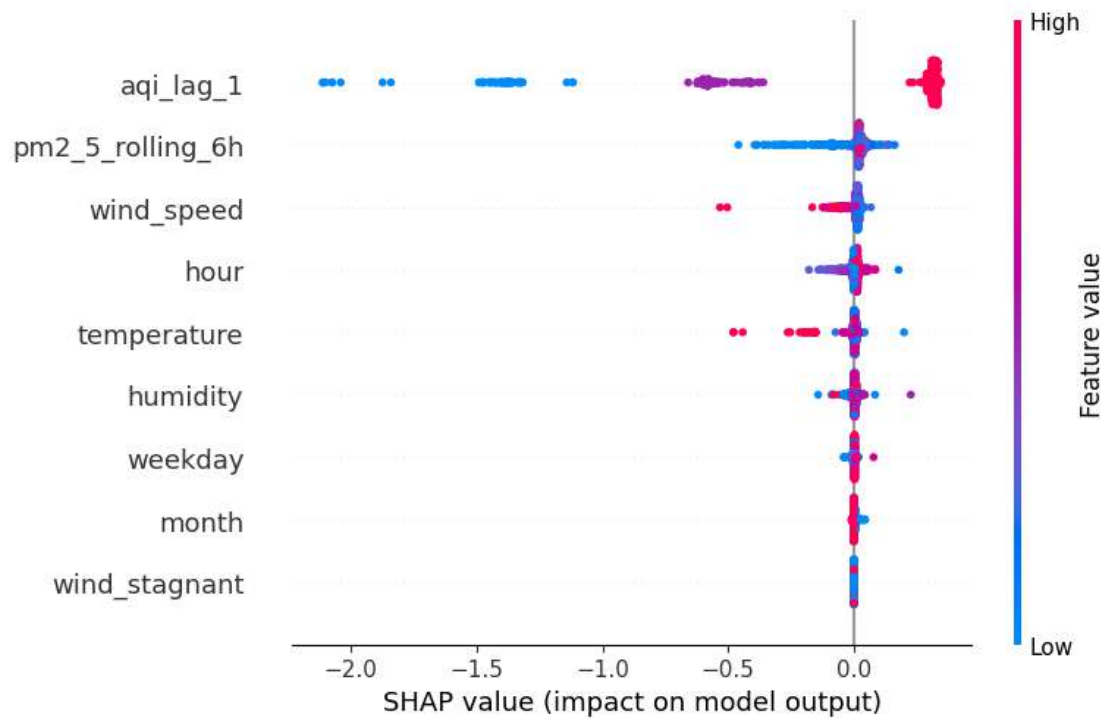
```
max_leaves=None, min_child_weight=None, missing=nan,  
monotone_constraints=None, multi_strategy=None, n_estimators=100,  
n_jobs=None, num_parallel_tree=None, ...)
```

Finished: Reading data from Hopsworks, using Hopsworks Feature Query Service
(0.52s)

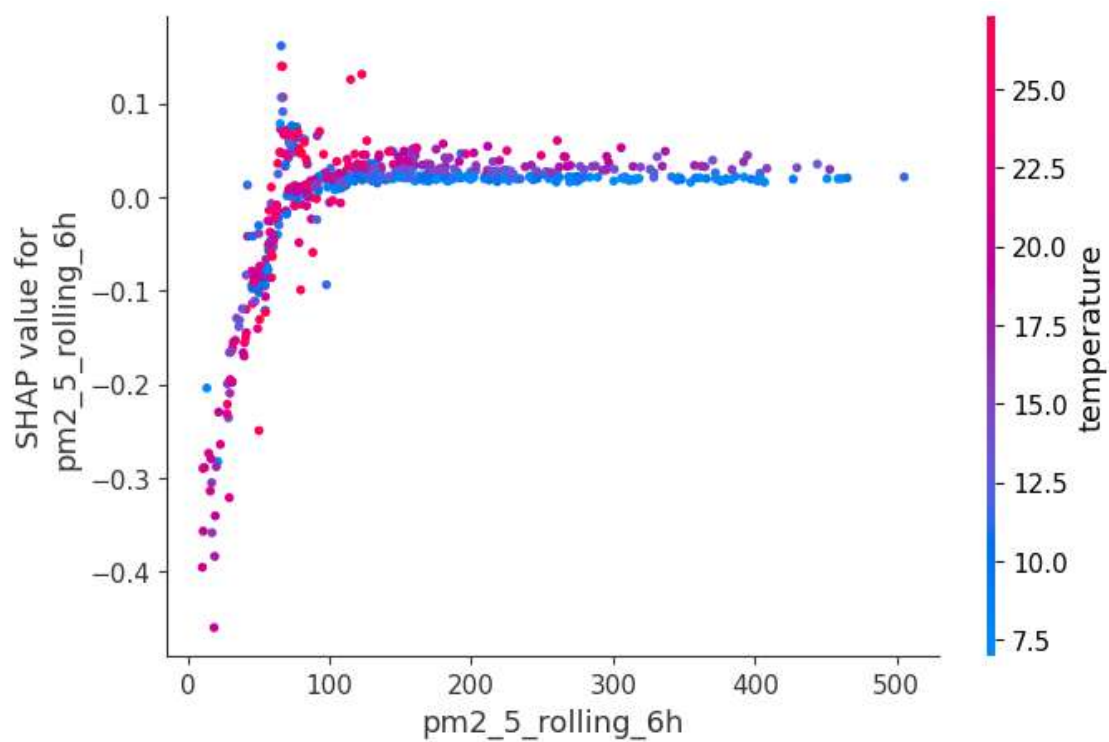
VersionWarning: Incremented version to `8`.

Features ready: ['temperature', 'humidity', 'wind_speed', 'hour', 'weekday',
'month', 'aqi_lag_1', 'pm2_5_rolling_6h', 'wind_stagnant']





<Figure size 640x480 with 0 Axes>



<IPython.core.display.HTML object>

SHAP analysis completed successfully in Colab!

<Figure size 640x480 with 0 Axes>